DERIVED HOM FROM DIFFERENTIAL GRADED BIMODULES IS FUNCTORIAL IN THE BIMODULE, AND QUASI-ISOMORPHISMS GIVE ISOMORPHISMS OF FUNCTORS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the property-(I) assembly of the transformation, the filtered reduction through the Rlim computation, the induction on p through the induced maps of short exact sequences, and the dual reduction concluding from the exactness of the dual are all retained verbatim.

Editorial note. The label lemma-functoriality-derived-restriction names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Computation of derived Hom by property (I) modules. Same chapter, Lemma lemma-derived-restriction. Identifies RHom(N, -) with Hom into differential graded B-modules with property (I), the starting point for both the transformation and its quasi-isomorphism analysis.
- Computing Rlim of filtered Hom systems and Hom into shifted duals. More on Algebra, Lemma more-algebra-lemma-compute-Rlim; Same chapter, Lemma lemma-hom-into-shift-dual-free. Lets the filtered system of quotients I/F_p I be read through complexes computing Rlim, and finally reduces each graded piece to the dual map on B-duals.

Source reference. Original reference 09LC: sources/stacks-dga/pinned-original-source.tex, source lines 3525–3525. Prerequisite pointer invoked at proof line(s) 5304; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09LI: sources/stacks-dga/pinned-original-source.tex, source lines 5272–5272. Prerequisite pointer invoked at proof line(s) 5311; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09K4: sources/stacks-dga/pinned-original-source.tex, source lines 2366–2366. Prerequisite pointer invoked at proof line(s) 5347; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 07KW: sources/stacks-chow/tags.txt, source lines 9671–9671. Prerequisite pointer invoked at proof line(s) 5326; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let $f : N \to N'$ be a homomorphism of differential graded (A, B)-bimodules. Then f induces a morphism of functors*

$$- \circ f : R\operatorname{Hom}(N', -) \longrightarrow R\operatorname{Hom}(N, -)$$

If f is a quasi-isomorphism, then $f \circ -$ is an isomorphism of functors.

Proof. Write $\mathcal{B} = \operatorname{Mod}^{dg}_{(B,\mathrm{d})}$ the differential graded category of differential graded B-modules, see Example 09LC. Let I be a differential graded B-module with property (I). Then $f \circ - : \operatorname{Hom}_{\mathcal{B}}(N', I) \to \operatorname{Hom}_{\mathcal{B}}(N, I)$ is a map of differential graded A-modules. Moreover, this is functorial with respect to I. Since the functors $R\operatorname{Hom}(N', -)$ and $R\operatorname{Hom}(N, -)$ are computed by applying $\operatorname{Hom}_{\mathcal{B}}$ into objects with property (I) (Lemma 09LI) we obtain a transformation of functors as indicated.

Assume that f is a quasi-isomorphism. Let $F_\bullet$ be the given filtration on I. Since $I = \lim I/F_pI$ we see that $\operatorname{Hom}_{\mathcal{B}}(N', I) = \lim \operatorname{Hom}_{\mathcal{B}}(N', I/F_pI)$ and $\operatorname{Hom}_{\mathcal{B}}(N, I) = \lim \operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$. Since the transition maps in the system I/F_pI are split as graded modules, we see that the transition maps in the systems $\operatorname{Hom}_{\mathcal{B}}(N', I/F_pI)$ and $\operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$ are surjective. Hence $\operatorname{Hom}_{\mathcal{B}}(N', I)$, resp. $\operatorname{Hom}_{\mathcal{B}}(N, I)$ viewed as a complex of abelian groups computes $R\lim$ of the system of complexes $\operatorname{Hom}_{\mathcal{B}}(N', I/F_pI)$, resp. $\operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$. See More on Algebra, Lemma 07KW. Thus it suffices to prove each

$$\operatorname{Hom}_{\mathcal{B}}(N', I/F_pI) \to \operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$$

is a quasi-isomorphism. Since the surjections $I/F_{p+1}I \to I/F_pI$ are split as maps of graded B-modules we see that

$$0 \to \operatorname{Hom}_{\mathcal{B}}(N', F_pI/F_{p+1}I) \to \operatorname{Hom}_{\mathcal{B}}(N', I/F_{p+1}I) \to \operatorname{Hom}_{\mathcal{B}}(N', I/F_pI) \to 0$$

is a short exact sequence of differential graded A-modules. There is a similar sequence for N and f induces a map of short exact sequences. Hence by induction on p (starting with $p = 0$ when $I/F_0I = 0$) we conclude that it suffices to show that the map $\operatorname{Hom}_{\mathcal{B}}(N', F_pI/F_{p+1}I) \to \operatorname{Hom}_{\mathcal{B}}(N, F_pI/F_{p+1}I)$ is a quasi-isomorphism. Since $F_pI/F_{p+1}I$ is a product of shifts of $A^\vee$ it suffice to prove $\operatorname{Hom}_{\mathcal{B}}(N', B^\vee[k]) \to \operatorname{Hom}_{\mathcal{B}}(N, B^\vee[k])$ is a quasi-isomorphism. By Lemma 09K4 it suffices to show $(N')^\vee \to N^\vee$ is a quasi-isomorphism. This is true because f is a quasi-isomorphism and $(\ )^\vee$ is an exact functor. □